

Welcome to MAT133!

Please find a seat and introduce yourself to your neighbours.

Lecture 1 **Welcome to MAT133!**

Big Questions

- Who are the instructors and lecture TAs?
- What is MAT133 about?
- Who are my classmates?

- My name is João Campos Vargas.
- Our class TA is Mila Zukanovic.
- We meet here on Mondays, Wednesdays, and Fridays from 2-3pm.
- Our Drop-in Study Hours will be on Wednesdays and Fridays from 3-4pm (starting next week).
- The course email is admin133@math.toronto.edu.

The MAT133 Instructor Team



Prof. Abhibhav
Garg



Prof. Adriano
Pacifico



Prof. Devin
Hensley



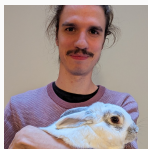
Prof. Jnaneshwar
Subhaschandra



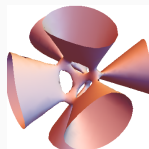
Prof. João
Campos Vargas



Prof. Pam
Sargent

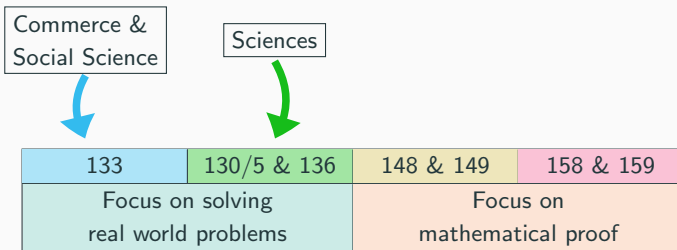


Prof. Luke
Volk



Prof. Simon
Xu

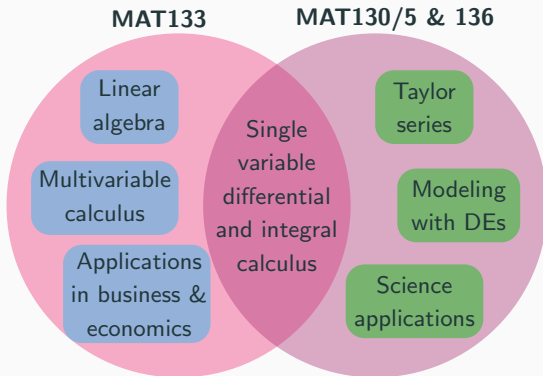
Calculus Course Comparison



- In each course, you'll experience both challenges and support (course code $\neq$ difficulty).
- Choose based on your goals and interests!
- Check the **Academic Calendar** for prerequisites.

MAT133 or MAT130/5 & 136?

- In either course, you'll build a solid foundation in single-variable calculus.
- MAT133 is more focused on real-world problems in business and economics.
- In MAT133, you'll learn additional mathematics to prepare you for further studies in Commerce and Economics.



What kinds of questions will we investigate?

Here are some examples of questions we will investigate:

**How can we
quantify income
inequality?**

**Can we predict
stock behaviour?**

**How can we use
data to inform our
decisions?**

Throughout the course, you'll get to work on real-world problems, using real data.

Overall Goals for MAT133

What mathematical skills are needed for your careers?



Flexibility in mathematical thinking



Problem-solving skills



Communication and collaboration



NOT memorization of formulas or algorithms

Important Dates

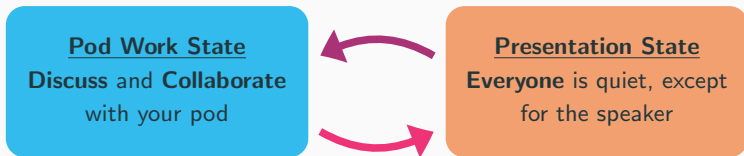
Lectures start on Wednesday Sept 9	Drop-in Study hours start on Monday Sept 14	Tutorials start on Thursday Sept 17
Wed. Sept 9	Mon. Sept 14	Thu Sept 17

Hives and Pods

- **Hives** meet in tutorials with their TA.
- **Pods** work together in lectures.
- **Pods** collaborate on projects.
- **Pods** present to their hives.



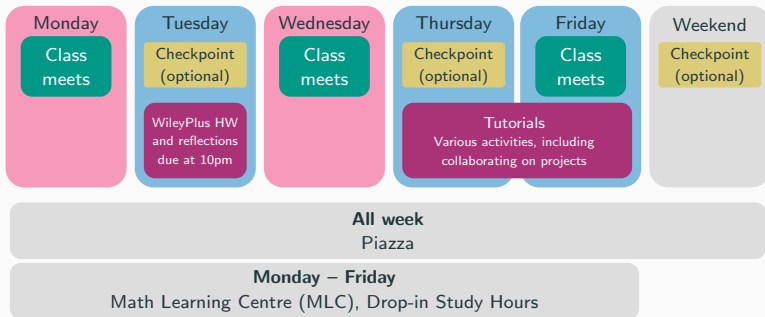
Two States of the MAT133 Classroom



Please be aware of which state the class is in.

If we have our **hands up**, we are asking for your attention.

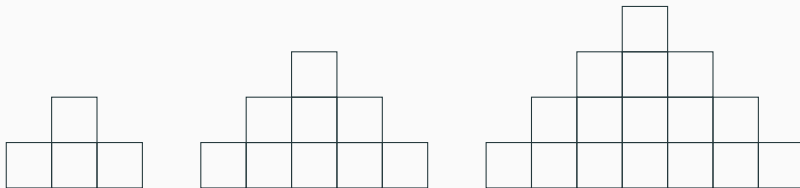
A Typical Week in MAT133



Please refer to the syllabus for all important dates!

Now to do some math!

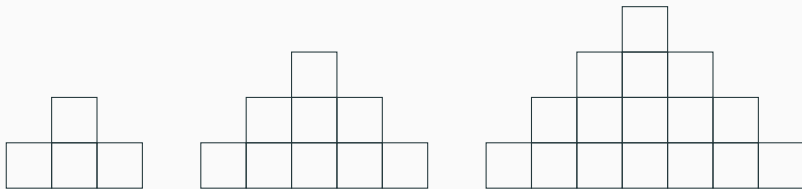
How do you see the shapes growing?



Think By Yourself

Visualize that this shape is growing. Where do the new blocks appear?

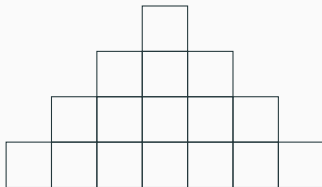
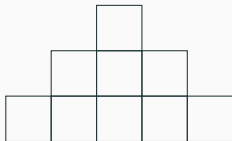
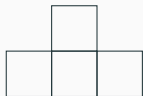
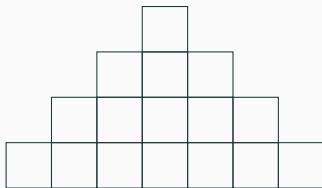
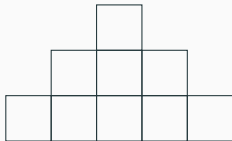
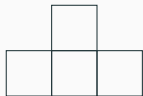
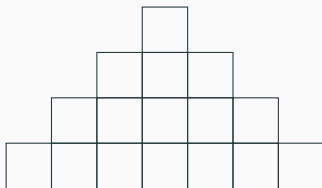
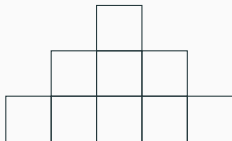
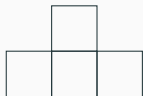
Mentally colour in the new blocks in each step and/or sketch on paper.

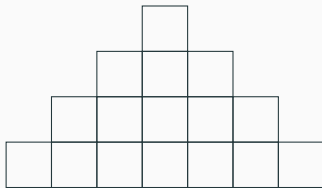
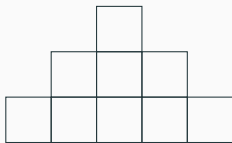
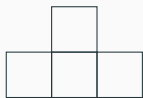
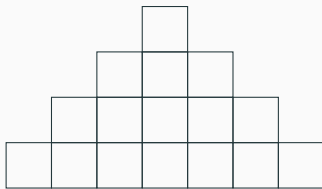
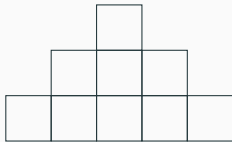
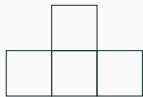
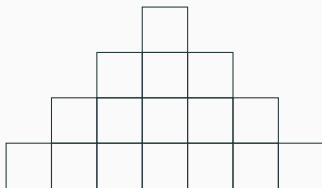
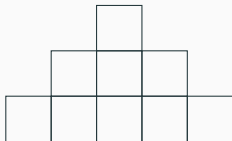
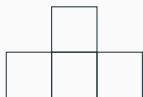


Discussion Question

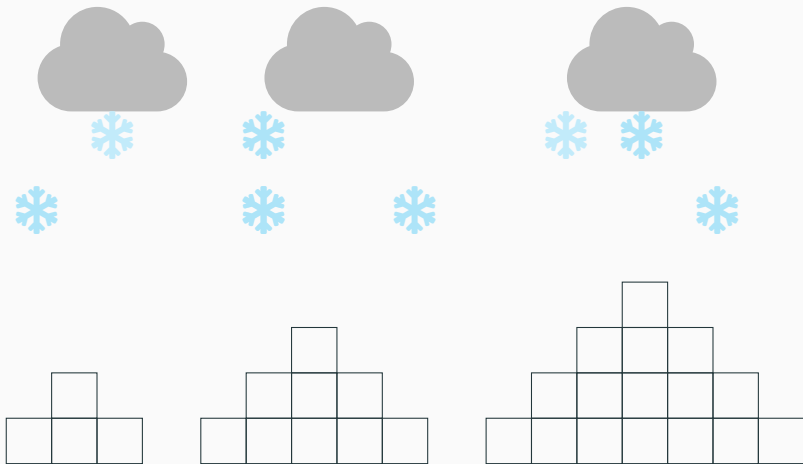
Take 5 minutes to discuss:

- (a) Introduce yourselves if you have not already done so!
- (b) Describe how you saw the shapes growing and share a sketch of your mental image of how they grow.

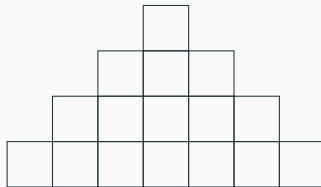
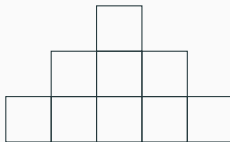
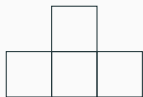




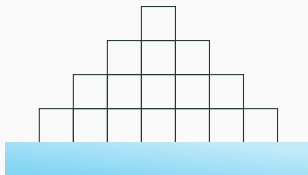
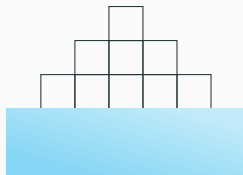
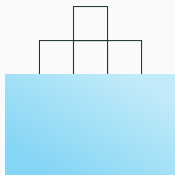
Imagine that it is snowing blocks...



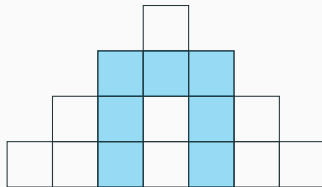
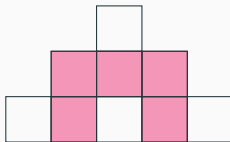
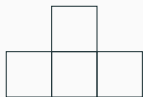
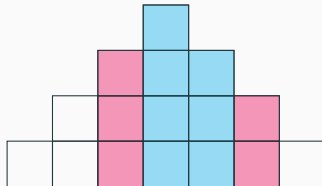
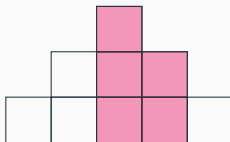
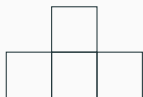
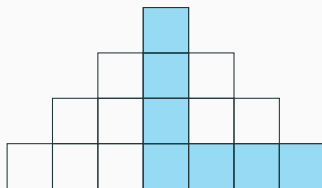
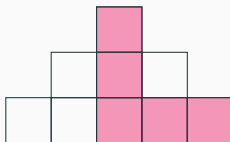
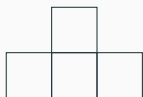
...or are the blocks rising out of the ground?



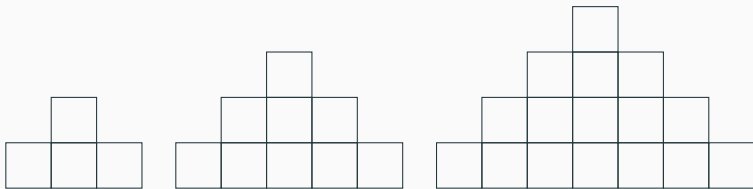
Or maybe they're being revealed as the snow melts!



A few creative ideas



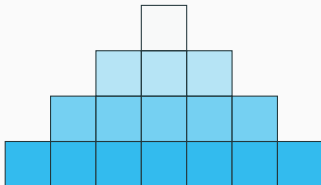
How is the area growing?



Discussion Question

How is the area growing? How many blocks are being added in each step?
Discuss.

What is the area of the shape?

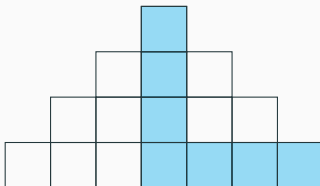


Imagine that the snow is melting, revealing the levels top to bottom.

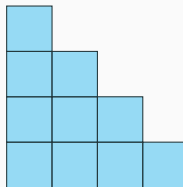
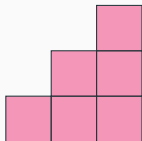
From this picture, we can see the area as a sum of

What if we use another picture?

Consider the shape with the large “L” highlighted:



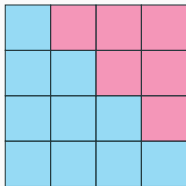
We can break the shape into two pieces:



Question 1

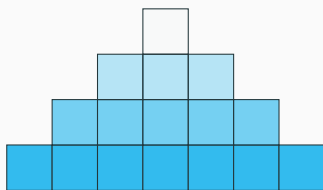
How can we combine these two pieces differently to more easily compute the area?

And stack one piece on top of the other:



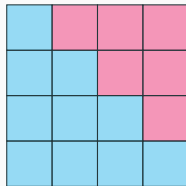
The area of the shape is

Putting our pictures together



$$1 + 3 + 5 + 7$$

=



$$4^2$$

We can see what happens when we let the shapes grow larger and larger.

For example, $1 + 3 + 5 + 7 + \dots + 99 =$.

This activity helps show that:

- Being **flexible** with different ways of thinking is important in math!
- **Collaboration** can help us to see different perspectives and learn new things!

Be sure to:

- Read the syllabus!
- If you haven't already done so:
 - Register in a tutorial on **ACORN**
 - Get your **UTORid**
- (No checkpoint for today's class!)

Have a great day!